Zhipeng Zhu

Data & AI Architect | Databricks Lakehouse | Enterprise AI
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SKILLS

DATA & AI PLATFORM
Databricks, Unity Catalog, Delta Lake
Spark / PySpark, Lakeflow
Vector Search, Metric Views, MLflow

GENAI & AGENTS
Genie Agents, Genie One
Knowledge Assistants, Databricks Apps
MCP, RAG, Agent Evaluation, HITL

DATA ENGINEERING
Python, R, SQL, Shell, Git
AWS (S3), Azure Data Factory
Hive, Medallion / ETL–ELT

GOVERNANCE & IDENTITY
Unity Catalog governance
RBAC, ABAC, Okta SSO, OBO identity

DOMAIN
Enterprise Agent architecture
Governed memory vs experiential RAG
HITL eval / write-back loops

EDUCATION

MSC, STATS / DATA SCIENCE
University of British Columbia
May 2022 | Vancouver, BC
Project & Co-op Double Track

BSC, PHYSICS / MATH & STATS
Ohio State University
May 2020 | Columbus, OH
Magna Cum Laude

WRITING / OSS

DISTPLYR & DISTIONARY
R / probaverse
Distribution objects (`dst_*`);
dplyr-style shift, mix, maximize.

ZHUZP98.GITHUB.IO
First-principles Agent notes.
Five-module enterprise Agent loop.
Databricks implementation map.

EXPERIENCE

DATA & AI ARCHITECT | BitMart
Feb 2025 - July 2026 | Remote

Platform: Architected warehouse→Databricks Lakehouse spanning 200+ datasets and 60+ users, with Unity Catalog governance and Okta SSO (RBAC/ABAC), with a ∼10-engineer delivery team. Established pipeline workflows that improved development pace ∼80% and cut maintenance/emergency events ∼90%.
Agents: Led the architecture and production deployment of Genie Agents, Knowledge Assistants, and Genie One Agents across 9 BUs (Finance, Operations, Marketing, Liquidity etc.). Business ad-hoc up to 20× faster; Finance fully automated with 90% process-hour savings; data-team ad-hoc volume down 50%.
Delivery: Translated requirements from Finance, Operations, Marketing, Liquidity into Databricks architectures; guided implementation and production rollout; ran adoption workshops and iterated from user feedback on AI Q&A. Presented outcomes to the company president and senior management. Orchestrated AWS Operations, Data Platform, Information Security, and Data Intelligence.
Integration: Delivered production Lark × Databricks integration—replaced Superset dashboard/CSV push with Databricks Workflow Jobs; synced Lark Wiki/Docs/Sheets into Unity Catalog as department Domain Context.

DATA ENGINEER | Budweiser APAC
Jul 2024 - Dec 2024 | Shanghai

Architecture: Designed Databricks middle-platform solutions; rebuilt ODS - DW - ADS dependencies; optimized clusters and Azure Data Factory ETL.
Delivery: Partnered with supply chain and sales to model DWS metrics and multi-environment governance for risk-order inspection and business insight.

SELECTED PROJECTS

Lark × Databricks Integration Solution
Aug 2026 - Present | Open Source (MIT License)

Governed identity and bidirectional workflows: **Genie Bot** (OBO Genie One in Lark); **Bridge Bot** (org tenant token, Lark↔Databricks); **Workspace Add-on** (OBO Genie in Docs/Base); **Lark MCP** for Genie Code/One.

Enterprise Databricks Lakehouse & Multi-Agent Platform
Feb 2026 - July 2026 | BitMart

Lakehouse & governance: S3→Unity Catalog (Medallion); finance data and algorithms (ERP/CRM); corporate docs as Context into UC and vector KBs; Okta SSO + RBAC + ABAC.
Agent orchestration: Dynamic Inputs → Knowledge/Memory → AI processing → Lark Execution → HITL eval/write-back—a closed, self-learning Agent system.

High-Frequency Liquidity & Effective Spread Platform
July 2025 - July 2026 | BitMart

Analytics: Designed Quant Liquidity metrics (quoted/effective spread, LOB resilience via recovery / mean-reversion / depth refill, adequacy A–E→L) with As-of Joins of trades vs 15-min LOB; 35% lower average effective spread on core pairs.
Platform: Migrated legacy hive_metastore tables to Unity Catalog and a subset of classic-compute jobs to Serverless; replanned liquidity ingestion (REST→full-book WebSocket, BM spot 184→1407), cutting cluster spend ∼50%; high-parallel execution took a 3-hour serial DAG to under 30 min.